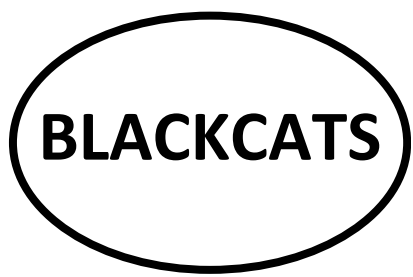


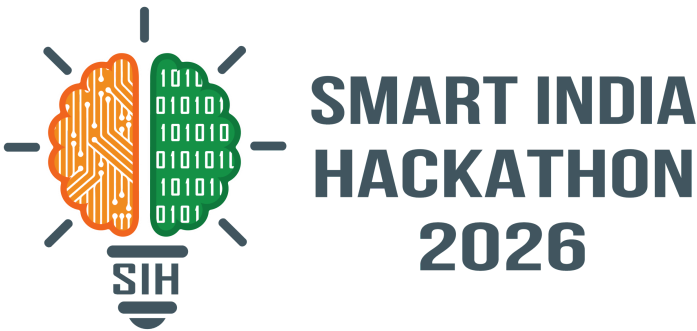
- Problem Statement ID – 26185
- Problem Statement Title- “Helmet mounted conformal antenna for tactical communications in urban CQB environments.”
- Theme- Robotics and Drones
- PS Category- Hardware
- Team ID- 007
- Team Name : BLACKCATS





Hi-CAS

HELMET-INTEGRATED CONFORMAL ANTENNA SYSTEM



THE PROBLEM → OUR SOLUTION

Traditional external antennas create a physical and RF trade-off in confined urban environments.

- 01
EXPOSED
HARDWARE

02
NON-CONFORMAL
FORM

03
RF PERFORMANCE
VARIATION

04
MULTI-BAND
CHALLENGE

External whip antennas protrude and snag on surroundings, interfering with operator movement.



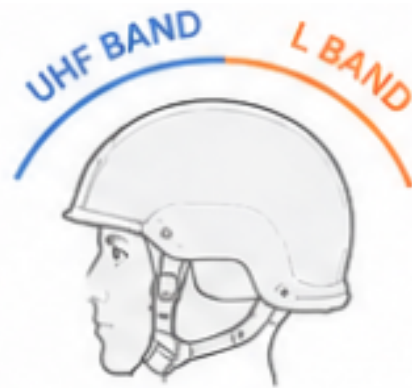
Rigid antennas do not follow helmet geometry, causing fit and performance issues.



Helmet curvature, proximity to human tissue, and environment affect antenna impedance and radiation.



One compact platform must reliably support and operate across UHF and L-band Frequencies.



OUR SOLUTION

Hi-CAS: Two independent low-profile conformal antenna arrays are integrated with the helmet surface and connected through a diplexer to enable reliable dual-band tactical communication

UHF ARRAY
200 – 450 MHz
Conformal array optimized for UHF band.

L-BAND ARRAY
1.2 – 1.5 GHz
Conformal array optimized for L-band.

DIPLEXER

UHF and L-band signals are combined/separated thorough a diplexer into a single RF interface

- CONFORMAL

Follows helmet curvature for seamless integration & comfort
- SINGLE-BAND
ARRAYS

Supports two required RF bands for tactical communication.
- LOW-PROFILE

Minimizes exposed hardware on the helmet surface & reduce snag risk.
- RETROFITTABLE

Designed for compatibility with existing helmet systems.
- PHASE SHIFTER

Optimizes signal coverage for reliable tactical communication.

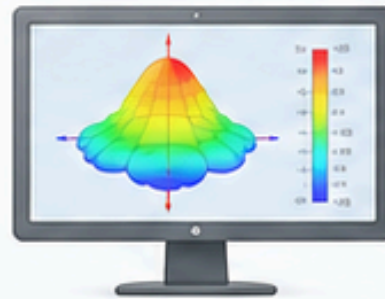
01 TECHNOLOGIES USED

HELMET &
BALLISTIC DESIGN

- Fusion 360
- Helmet geometry
- Ballistic simulation

ANTENNA &
EM SIMULATION

- CST Studio Suite
- UHF & L-band antenna design
- EM / radiation analysis



HARDWARE

- UHF Array — 200–450 MHz
- L-band Array — 1.2–1.5 GHz
- Flexible PCB / copper traces
- 3-port RF diplexer
- Phase-control network
- RF interconnects
- Radio / transceiver



02 IMPLEMENTATION FLOW

01



REQUIREMENTS

Frequency •
Coverage •
Helmet constraints

02

HELMET +
ARRAY DESIGN

Fusion 360

03

ANTENNA & EM
SIMULATION

CST Studio Suite

04

ARRAY
OPTIMIZATION

Spacing • Curvature •
Radiation pattern

05



FABRICATION

Flexible PCB +
RF interconnects

06

HELMET
INTEGRATION

UHF + L-band +
Diplexer

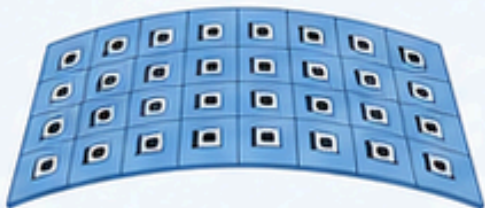
07



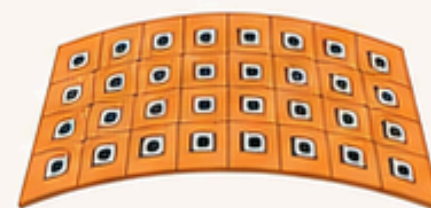
VALIDATION

RF Performance +
Ballistic Performance

03 WORKING CONCEPT

UHF ARRAY
200 – 450 MHz

3-PORT DIPLEXER

L-BAND ARRAY
1.2 – 1.5 GHz

RADIO / TRANSCEIVER



04 SIMULATION & VALIDATION



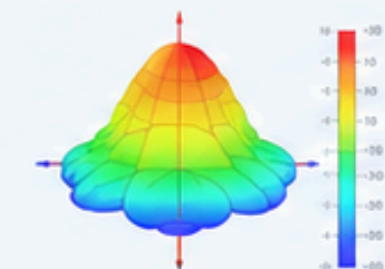
FUSION 360

Helmet + ballistic design



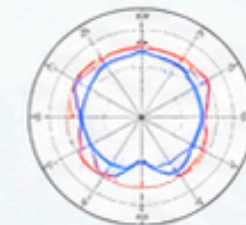
CST STUDIO SUITE

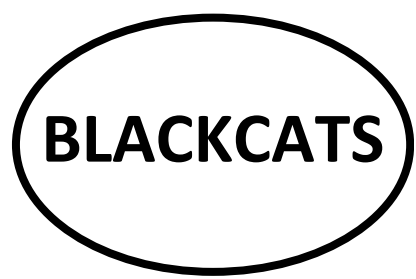
Antenna + electromagnetic
simulation



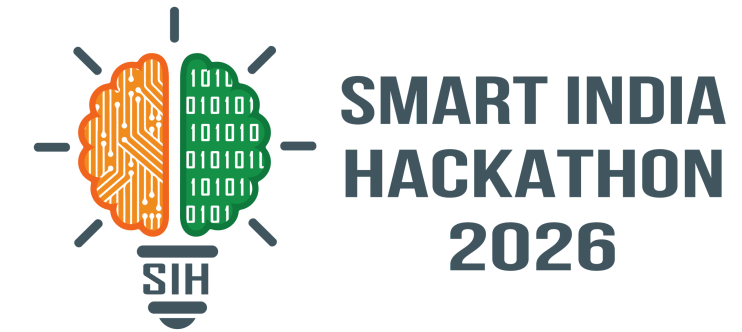
VALIDATION

- S11 / VSWR
- Radiation pattern
- Gain
- RF isolation
- Ballistic performance





Feasibility + Engineering Challenges



Core technologies exist.

Innovation is in integration and validating under realistic conditions.

CURVATURE

Bending alters electrical length and EM coupling.

Measure resonant frequency, S11, VSWR, bandwidth.



TEMPERATURE

Environmental temperature variations impact material and antenna behavior.

Measure RF performance before and after temperature cycling.



HEAD PROXIMITY

Operator's head changes the EM environment around antenna.

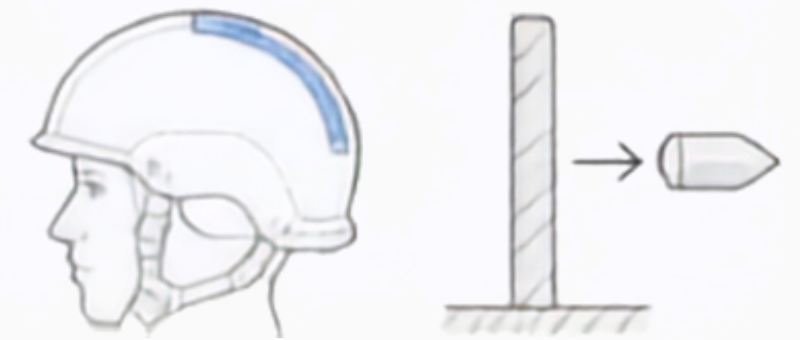
Measure impedance, efficiency, and radiation characteristics.



STRUCTURAL INTEGRATION

Antenna must not compromise helmet functionality or protection.

Measure integration compatibility and structural/ballistic validation requirements (no certification claimed).



MITIGATION + VALIDATION SEQUENCE

01

DESIGN



02

SIMULATE



03

PROTOTYPE



04

BEND TEST



05

TEMPERATURE TEST



06

RF VALIDATION

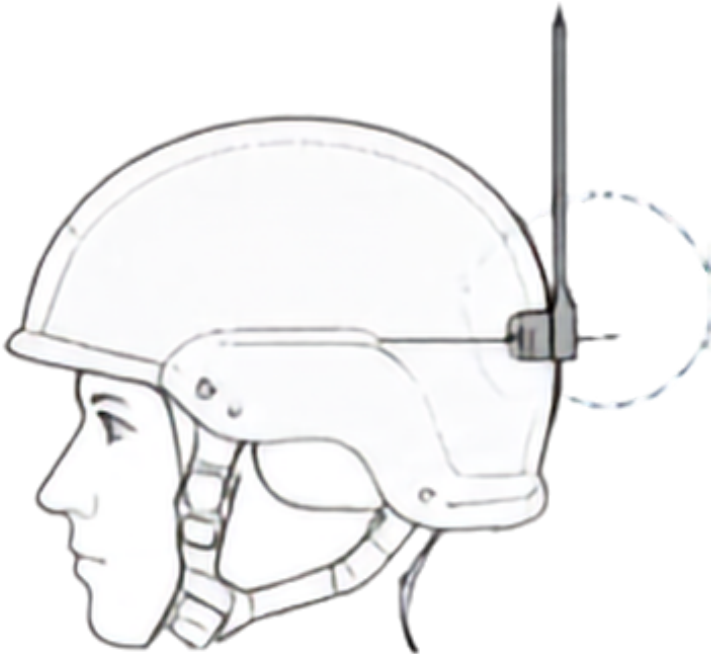


07

STRUCTURAL
VALIDATION

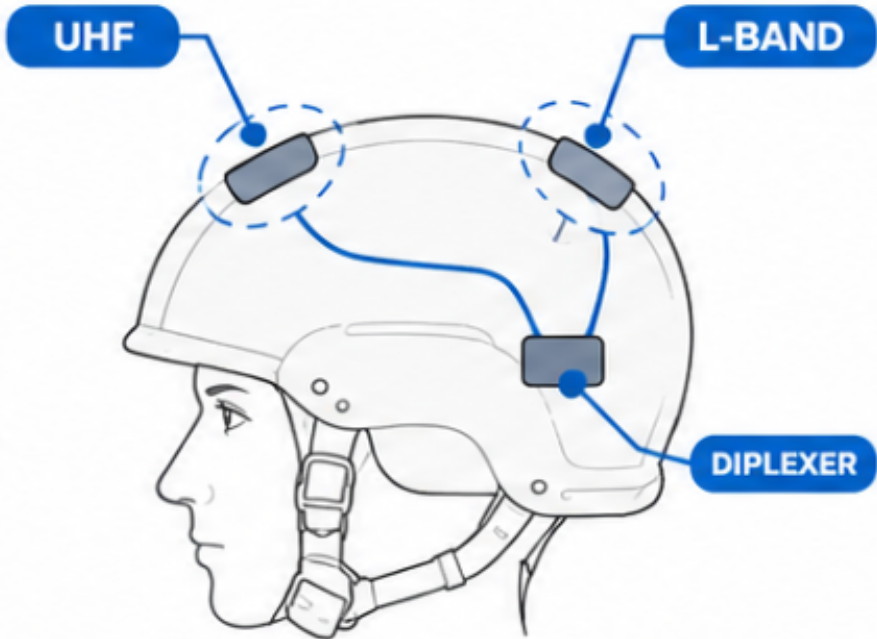
FROM EXTERNAL ANTENNA
TO INTEGRATED COMMUNICATION

CONVENTIONAL EXTERNAL ANTENNA



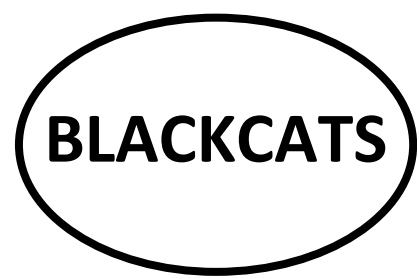
Protruding hardware
Potential snag points
Physically exposed
Geometry independent of helmet

HI-CAS

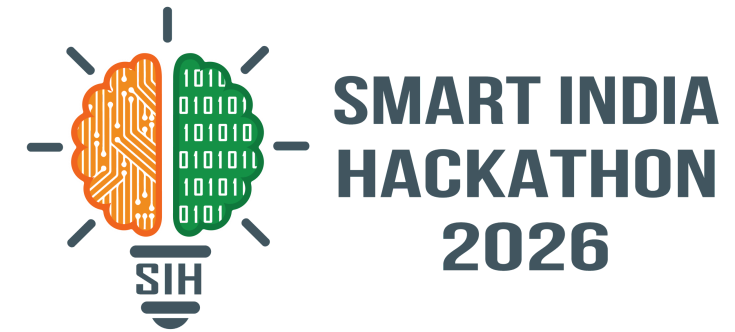


Separate UHF & L - band arrays
UHF: 200 - 450 MHz
L - band : 1.2 - 1.5 Ghz
Diplexer combines both bands
Integrated, helmet-conformal RF structure

IMPACT AREAS	
OPERATIONAL	Reduced physical obstruction, improved operator mobility.
RF PERFORMANCE	Independent band-optimized arrays with diplexer combining
USER SAFETY	Fewer exposed antenna parts reduce snag and injury risks.
SCALABILITY	Modular UHF/L-band architecture for broad helmet and radio compatibility.
POTENTIAL APPLICATIONS	
NSG / ATS teams, specialized tactical units, other helmet-based communication systems, subject to validation and procurement.	



RESEARCH AND REFERENCES



- GitHub repository:

https://github.com/Sriharshithprasadns456/blackcats_sih26